

# Vidisha Deshpande

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## Education

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**State University of New York, Stony Brook** | *Masters in Data Science* Aug 2025 – May 2027  
◦ **GPA: 3.78**    Coursework: Natural Language Processing, Data Analysis, Statistics, Probability  
**University of Pune** | *Bachelors in Engineering* Aug 2021 – May 2025  
◦ **GPA: 9.23/10**    Coursework: Data Analytics, Database Systems, Deep Learning

## Experience

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**BMC Helix** Jun 2026 – Aug 2026  
*Data Science Intern*

- Built a conversational ITSM agent on BMC Helix Agent Studio that translates unstructured natural-language input into structured tickets, replacing traditional form-based intake with LLM-driven field inference and deterministic priority assignment.
- Prototyped natural-language ticket search using LLM-generated keyword expansion with lightweight text matching.
- Evaluated a production conversational AI agent in an enterprise IT portal through structured adversarial testing, identifying and documenting failure modes in intent routing and multi-turn conversation state handling.

## Projects

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**FactGraph: Knowledge Graph-Grounded Hallucination Detection for LLM Annotations** [github.com/vidi-deshp12](https://github.com/vidi-deshp12) [🔗](#) Mar 2026 – May 2026  
*Python, spaCy, Neo4j, LLaMA 3.1, MiniLM, DeBERTa, Qwen2.5*

- Reduced LLM annotation errors by **62%** across 7,724 FEVER dev claims by building a KG-grounded verification pipeline: LLaMA annotation → spaCy triple extraction → Neo4j KG retrieval → DeBERTa/Qwen2.5 NLI verification.
- Boosted NEI (Not Enough Info) recall **4.9×** from 0.17 to 0.83 by grounding ambiguous claims in structured Wikidata evidence rather than relying on parametric LLM knowledge.
- Constructed a FEVER-targeted Wikidata Knowledge Graph of 3,990 triples across 700 entities and engineered a three-route Neo4j retrieval system (exact match → property lookup → MiniLM semantic fallback) with relation-direction correction and object normalization to handle KG schema mismatches in natural-language claims.
- Compared three verification configurations, quantifying accuracy–recall tradeoffs, demonstrating KG retrieval as a targeted correction layer over overconfident LLM outputs.

**Real-Time Image Captioning for the Visually Impaired** [github.com/vidi-deshp12](https://github.com/vidi-deshp12) [🔗](#) Aug 2024 – May 2025  
*Final Year Thesis | Python, PyTorch, Flask, Hugging Face*

- Reduced end-to-end inference latency by **44%** (9s → 3.2s) for visually impaired users by parallelizing CLIP image encoding and EasyOCR scene-text extraction, enabling real-time audio caption delivery on mobile.
- Designed a parameter-efficient ClipCap-style captioning model mapping **512-d CLIP embeddings** to **GPT-2 prefix tokens** via an **8M-parameter MLP** with both backbones frozen, trained on 20K+ VizWiz accessibility images, keeping compute cost low without sacrificing caption quality.
- Deployed fine-tuned model to Hugging Face Hub and built a fully functional full-stack prototype (React Native frontend + Flask REST inference backend serving real-time captioning requests), validated end-to-end under real-world conditions including low-light, motion blur, and occlusion.
- Co-authored peer-reviewed publication in SmartCom 2025 based on this work.

## Publications

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[Empowering Vision: A Survey on Assistive Technologies for the Visually Impaired](#) [🔗](#) (**SmartCom 2025**) (peer-reviewed)  
[Few-Shot Learning in Vision-Language Models: A Survey](#) (**CVR 2025**) (peer-reviewed)

## Technical Skills

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**Languages:** Python, SQL  
**Databases:** PostgreSQL, Neo4j  
**ML/DL Frameworks:** PyTorch, TensorFlow, Keras, Hugging Face Transformers  
**Tools & Platforms:** Git, Docker, Flask, spaCy  
**Specializations:** NLP, Computer Vision, Vision-Language Models, LLM Evaluation, Multimodal Learning